

How The IoT Has EVOLVED From M2M COMMUNICATION

The IoT is a network of devices connected to ISPs or WAN/MAN through an M2M gateway



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The Internet of Things (IoT) emerged from the earlier work on machine-to-machine (M2M) communication. M2M was the beginning of taking data from devices/sensors and feeding it to the Internet through M2M gateways, for data processing servers and more. It is called M2M because there is no human intervention required, right from data generation to data processing.

According to Hyunmi Yang, GSMA chief strategy officer, mobile networks are the platform upon which the M2M industry is being built. Mobile operators are at the forefront in shaping new business models that are driving this exciting market forwards.

Essentially, the IoT is a network of devices—called capillary network—connected to ISPs or WAN/MAN through an M2M gateway. The whole enablement of IT is managed through remote connectivity.

Researchers took the concept of

things that can create or generate data, and transformed it as the IoT. Some also referred to it as the Internet of Everything (IoE).

In literary terms, the IoT is the expansion of connectivity to things that go beyond human assistance. It is about connectivity of devices, with zero human interaction, to far-reaching data centres. It requires radios, devices and sensors for low-range and low-power connectivity. Then, there are technologies like BLE, LoRa, Zigbee, Sigfox and LPWAN that support the IoT.

The IoT is linked to Big Data and analytics. It can help and be attached to artificial intelligence (AI) and machine learning, too.

The 3rd Generation Partnership Project (3GPP) has standards such as LTE-M, NB-IoT and GSM-specific technological tweaks to cater to certain issues. But there are no specific standards for the IoT as a holistic system.

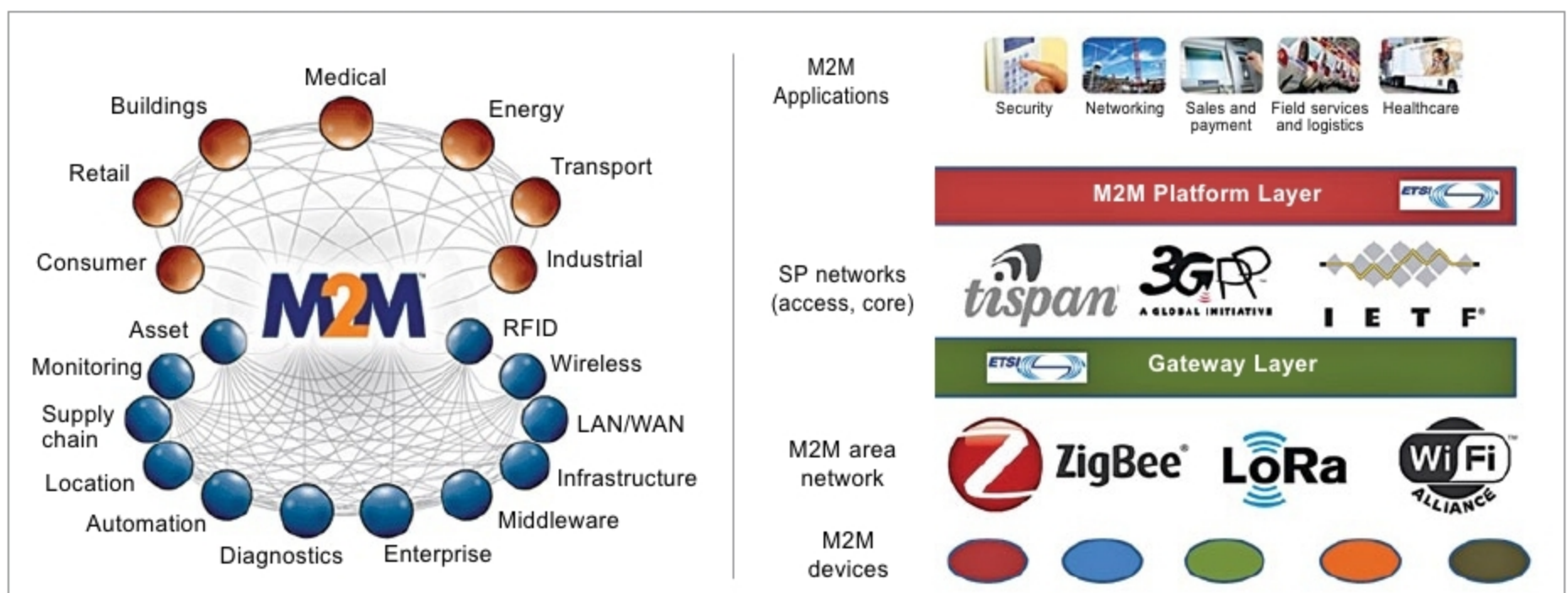


Fig. 1: Concept of M2M communication